

Local Novelty Compression and Exploratory Saturation in Recursive Inferential Systems

The OCES Core Operator Framework [†]

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[†] Version 3. All results: $d=16$, 50 seeds. Cross-dimensional validation ($d \in \{8, 32, 64\}$): companion paper.

Abstract

We introduce the Operational Constrained Exploratory Systems (OCES) core operator framework for measuring exploratory saturation in recursive inferential systems under bounded representational geometry. Five operators are derived from first principles: local novelty functional N_t , compression field C_t , restructuring persistence R_t , accessibility entropy A_t , and participation ratio PR_t . These are unified into a field operator $\Gamma_t = C_t \cdot A_t \cdot R_t \cdot \Delta PR(t)$ with a formal saturation convergence theorem, and augmented by a Joint Saturation Certificate (Definition 2) that resolves the structural ambiguity of the multiplicative form. Across 50 independent seeds ($d=16$, 50 000 steps), behavioural novelty exhibits *phase-structured local compression*: rising to a peak $N_{\text{peak}}=0.151$ at 25k steps, then compressing to $N_{\text{final}}=0.089 \pm 0.019$ (95% CI [0.084, 0.095]), while functional utility rises monotonically ($U=0.540$). The coverage-novelty anti-correlation is $r=-0.513$ ($p=0.0002$, two-tailed permutation, $n=10\,000$; 95% CI $[-0.62, -0.38]$), identifying representational occupancy as the geometric driver. $\Gamma_t \rightarrow 0$ in 100% of seeds, and the Joint Certificate ($C_t \rightarrow 0 \wedge \Gamma_t \rightarrow 0$) is satisfied in all 50 seeds, providing the first operationally measurable saturation classifier for recursive exploratory dynamics. Code and data are available at [https://github.com/\[repo\]/OCES](https://github.com/[repo]/OCES).

Keywords: exploratory saturation; novelty compression; recursive inferential systems; participation ratio; open-ended learning.

1 Introduction

Whether recursively self-modelling architectures can sustain unbounded behavioural novelty is a foundational open problem in artificial life and adaptive AI. Exploration collapse shows up everywhere you look: RL agents narrow to reward-maximising policies [Sutton and Barto, 2018]; language models overfit to their training distri-

butions [Brown et al., 2020]; creative systems start repeating themselves after long enough runs [Lehman and Stanley, 2011].

Existing frameworks tackle this through behavioural novelty optimisation (Novelty Search [Lehman and Stanley, 2011]), quality-diversity archiving (MAP-Elites [Mouret and Clune, 2015]), and curiosity-driven prediction-error bonuses [Pathak et al., 2017]. They are designed to *drive* exploration. None of them provide a principled *measurement* layer for quantifying when, and how badly, novelty generation saturates asymptotically. That is what we build here.

The key distinction. OCES is a *measurement and diagnostic architecture*, not a search algorithm. It does not compete with MAP-Elites or Novelty Search — it tells you which regime those algorithms are operating in.

Contributions:

1. Five core operators with complete formal proofs (§3).
2. Master equation Γ_t with saturation convergence theorem.
3. Joint Saturation Certificate resolving Γ_t ambiguity (Definition 2; **new in v2**).
4. Phase-structured compression validated across 50 seeds.
5. Coverage-novelty anti-correlation with correct two-tailed statistics (**corrected in v2**).
6. Full 7-condition causal ablation with bootstrap CIs (**extended in v3**).

Scope. All results are for $d=16$. Cross-dimensional validation ($d \in \{8, 32, 64\}$) is in the companion paper.

2 Related Work

Novelty Search and QD. Lehman and Stanley [2011] showed that optimising for k-NN archive distance can outperform objective search. Our N_t functional is directly analogous — the difference is that we use it to *measure*, not to optimise. MAP-Elites [Mouret and

Clune, 2015] provides structured archive occupancy; our negative coverage-novelty correlation is consistent with its geometric diversity pressure. Neither framework provides a measurement layer for asymptotic saturation.

Intrinsic motivation. Prediction-error bonuses [Pathak et al., 2017] sustain exploration where error stays non-zero, but do not address geometric occupancy saturation. Our results suggest representational occupancy imposes limits that prediction-error bonuses simply cannot circumvent.

Open-ended evolution. Bedau [2003] and Taylor et al. [2016] sought objective measures of ongoing novelty without a framework for asymptotic behaviour. The participation ratio PR_t draws on statistical physics [Bell and Dean, 1970]; its coupling with novelty compression is novel to this work.

3 Core Operator Framework

3.1 Problem Setup

Definition 1 (State space). $\mathcal{M} = \{\Phi \in \mathbb{R}^d : \|\Phi\| \leq 1.0\}$, with $d=16$ and Euclidean metric $d_g(x, y) = \|x - y\|_2$.

Each ontology maintains state $\Phi_i \in \mathcal{M}$ and an evolvable two-layer inference law evolved via natural Evolution Strategies [Wierstra et al., 2014]:

$$f_\theta(\Phi, c) = \tanh(W_2 \tanh(W_1[\Phi; c] + b_1) + b_2). \quad (1)$$

Research question. Under bounded representational geometry and recursive self-modelling, does $\mathbb{E}[N_t]$ compress asymptotically? If so, which operators drive it?

3.2 Notation Summary

Table 1 lists all symbols used throughout the paper.

Table 1: Summary of notation.

Symbol	Definition	Eq.
$\mathcal{M}$	State space $\{\Phi \in \mathbb{R}^d : \ \Phi\ \leq 1\}$	Def. 1
d_g	Euclidean metric $\ x - y\ _2$	Def. 1
$\mathcal{A}_t$	Behavioural archive at time t	—
$N_t(x)$	Local novelty functional	(2)
C_t	Compression field	(3)
R_t	Restructuring persistence	(4)
A_t	Accessibility entropy	(6)
PR_t	Participation ratio	(7)
Γ_t	Master field operator (mult.)	(8)
Γ_t^{add}	Master field operator (add.)	(9)
S_i	Causal sensitivity score	(10)
Def. 2	Joint Saturation Certificate	—

3.3 Operator 1: Local Novelty Functional N_t

$$N_t(x) = \frac{1}{k} \sum_{i=1}^k d_g(x, \text{NN}_i(x, \mathcal{A}_t)), \quad k=5. \quad (2)$$

The compression rate $\gamma(t) = -d/dt \mathbb{E}[N_t]$: $\gamma > 0$ is compression; $\gamma < 0$ is expansion.

Theorem 1 (Boundedness). $0 \leq N_t(x) \leq d_{\max}$ for all x, t . $N_t(x) = 0$ iff x coincides with $\geq k$ archive members.

Proof. Lower bound: $d_g \geq 0$ (metric axiom); mean of non-negatives is non-negative. Upper bound: each summand $d_g(x, \text{NN}_i) \leq d_{\max}$, so $N_t \leq (1/k) \cdot k \cdot d_{\max} = d_{\max}$. Zero condition: positive-definiteness gives $d_g(x, y) = 0 \Leftrightarrow x = y$. $\square$

3.4 Operator 2: Compression Field C_t

$$C_t = 1 - \frac{\mathbb{E}[N_{t+w}]}{\mathbb{E}[N_t]}, \quad w = 1000 \text{ steps}. \quad (3)$$

$C_t > 0$: compression; $C_t < 0$: expansion; $C_t = 0$: steady state. The signed formulation is load-bearing — clipping to $[0, 1)$ destroys expansion-phase information and artificially inflates N_{peak} estimates.

Theorem 2 (Saturation criterion). $\lim_{t \rightarrow \infty} C_t = 0$ iff $\mathbb{E}[N_t] \rightarrow \text{constant}$. C_t is invariant to global scaling of $\mathbb{E}[N_t]$.

Proof. If $\mathbb{E}[N_t] \rightarrow L > 0$: $\lim C_t = 1 - L/L = 0$. Conversely, $C_t \rightarrow 0$ implies $\mathbb{E}[N_{t+w}]/\mathbb{E}[N_t] \rightarrow 1$; by Stolz-Cesàro under mild regularity, $\mathbb{E}[N_t] \rightarrow L$. Scale-invariance: multiplying $\mathbb{E}[N_t]$ by α cancels in the ratio. $\square$

3.5 Operator 3: Restructuring Persistence R_t

$$R_t = \frac{|O_{t+w} \setminus O_t|}{|O_{t+w}|}, \quad \delta = 0.15 \text{ (grid resolution)}. \quad (4)$$

O_t is the set of occupied grid cells at time t . $R_t = 0$ when no new cells are discovered; $R_t > 0$ certifies ongoing representational renewal.

Theorem 3 (R_t saturation). $R_t \in [0, 1]$ for all t . Under eventually periodic exploration, $\lim_{t \rightarrow \infty} R_t = 0$. $R_t \geq \varepsilon > 0$ iff exploration is aperiodic.

Proof. Range. $|O_{t+w} \setminus O_t| \subseteq O_{t+w}$ by set inclusion, so numerator $\leq$ denominator; both non-negative; $R_t \in [0, 1]$.

Periodic limit. Suppose exploration is eventually periodic with period T . Then $\exists T_0$ such that $\forall t \geq T_0$: $O_{t+T} = O_t$. For window w a multiple of T : $O_{t+w} \setminus O_t = \emptyset$, giving $R_t = 0$.

Persistence. $R_t \geq \varepsilon$ requires $|O_{t+w} \setminus O_t| \geq \varepsilon|O_{t+w}|$ at every window — a fraction ε of current cells are always genuinely new, which is precisely the definition of aperiodicity. $\square$

Remark 1. $R_t > 0.05$ at saturation across all 50 seeds (Table 4) certifies structural aperiodicity by Theorem 3. This is a structural certificate, not a soft indicator.

3.6 Operator 4: Accessibility Entropy A_t

$$G_t = (V_t, E_t), \quad E_t(i, j) = \mathbf{1}[\|x_i - x_j\| < r_t], \quad (5)$$

$$A_t = -\sum_i p_i(t) \log p_i(t), \quad p_i = \frac{|C_i(t)|}{|\mathcal{A}_t|}, \quad (6)$$

where $C_i(t)$ are connected components of G_t .

Theorem 4 (Accessibility bounds and saturation convergence). $0 \leq A_t \leq \log |\mathcal{A}_t|$. Under archive densification at saturation: $A_t \rightarrow 0$.

Proof. Bounds. A_t is the Shannon entropy of $p_i(t) = |C_i(t)|/|\mathcal{A}_t|$ over connected components. By standard information theory, entropy is maximised when all components have equal size, giving $A_t \leq \log |\mathcal{A}_t|$. Non-negativity follows from $p_i(t) \in [0, 1]$ and the convention $0 \cdot \log 0 = 0$.

Saturation convergence. As the archive densifies at saturation, the connectivity radius r_t ensures that for any two archived states $x, y \in \mathcal{A}_t$, $d_g(x, y)$ eventually falls below r_t . All vertices become connected in a single component: $|C_i(t)| \rightarrow |\mathcal{A}_t|$ for one i , and $|C_j(t)| \rightarrow 0$ for $j \neq i$. Hence $p_i(t) \rightarrow 1$ for the dominant component and $p_j(t) \rightarrow 0$ for all others. Substituting: $A_t \rightarrow -(1 \cdot \log 1 + 0 \cdot \log 0 + \dots) = 0$. $\square$

Remark 2 (Adaptive connectivity radius). For the $d=16$ results reported here, the fixed radius $r=0.3$ is sufficient. For cross-dimensional generalisation ($d \in \{8, 32, 64\}$), the radius must scale as $r(d) = 0.3/\sqrt{d/16}$ to maintain consistent neighbourhood density. This scaling follows from the volume of a d -ball: $V_d(r) \propto r^d$, so constant local density requires $r_d \propto d^{-1/2}$. The adaptive version is implemented and validated in companion Paper II.

3.7 Operator 5: Participation Ratio PR_t

$$\text{PR}_t = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}, \quad (7)$$

where λ_i are eigenvalues of $\text{Cov}(\text{states})$. $\text{PR}_t = 1$: single-axis collapse; $\text{PR}_t = d$: fully distributed.

Theorem 5 (PR_t bounds). For any PSD covariance matrix with normalised eigenvalues $s_i = \lambda_i / \sum_j \lambda_j$ ($\sum_i s_i = 1$, $s_i \geq 0$):

$$1 \leq \text{PR}_t \leq \text{rank}(\text{Cov}) \leq d.$$

Proof. Let $m = |\{i : s_i > 0\}| = \text{rank}(\text{Cov})$. By Cauchy-Schwarz: $(\sum_i s_i)^2 \leq (\sum_i 1^2) \cdot (\sum_i s_i^2) = m \cdot \sum_i s_i^2$. Since $\sum_i s_i = 1$: $1 \leq m \cdot \|s\|^2$, hence $\text{PR}_t = 1/\|s\|^2 \leq m \leq d$.

For the lower bound: $\|s\|^2 \leq \max_i s_i \cdot \sum_i s_i = \max_i s_i \leq 1$, so $\text{PR}_t = 1/\|s\|^2 \geq 1$. Equality $\text{PR}_t = 1$ occurs iff one eigenvalue dominates (rank-1). Equality $\text{PR}_t = d$ occurs iff all eigenvalues are equal ($s_i = 1/d$). $\square$

Remark 3 (Empirical collapse). In our experiments, PR_t collapsed from 7.34 ± 0.52 (5k steps) to 2.91 ± 0.38 (50k steps) — a loss of ≈ 4.43 effective dimensions, a 60% reduction. The accessible occupancy graph remained fully connected (5/5 partitions reachable) throughout, confirming dimensional contraction is axis-selective, not topological fragmentation.

3.8 Master Equation Γ_t

$$\Gamma_t = C_t \cdot A_t \cdot R_t \cdot \Delta \text{PR}(t). \quad (8)$$

Regimes: $\Gamma_t > 0$ (adaptive); $\Gamma_t \approx 0$ (saturated); $\Gamma_t < 0$ (collapse). To address multiplicative fragility when any single factor temporarily drops near zero, we track:

$$\begin{aligned} \Gamma_t^{\text{add}} = & 0.414 C_t + 0.138 A_t \\ & + 0.138 R_t + 0.107 \Delta \text{PR}(t) + 0.203 \varepsilon_{\text{res}}, \end{aligned} \quad (9)$$

where weights come directly from the causal sensitivity scores S_i (Section 6, Table 5): $S_{\text{archive}}=0.414$, $S_{\text{occupancy}}=0.138$, $S_{\text{noise}}=0.107$, with residual 0.203 capturing coupling, utility, and higher-order interactions. The rounded weights (0.30, 0.25, 0.25, 0.20) used in prior versions are approximations; Eq. (9) gives the exact empirically-grounded form.

$$S_i = \frac{|\Delta \Gamma_{t,i}^{\text{add}}|}{\sum_j |\Delta \Gamma_{t,j}^{\text{add}}|}. \quad (10)$$

Theorem 6 (Saturation convergence). Under $\mathbb{E}[N_t] \rightarrow L > 0$: $\lim_{t \rightarrow \infty} \Gamma_t = 0$. $C_t \rightarrow 0$ forces $\Gamma_t \rightarrow 0$ regardless of A_t , R_t , ΔPR .

Proof. From Theorem 2: $C_t \rightarrow 0$. $A_t \in [0, \log |\mathcal{A}_t|]$ (Theorem 4), $R_t \in [0, 1]$ (Theorem 3), $|\Delta \text{PR}| \leq d-1$. Hence $|\Gamma_t| \leq |C_t| \cdot (d-1) \cdot \log |\mathcal{A}_t| \rightarrow 0$. $\square$

Remark 4 (Principal ambiguity). The converse fails: $\Gamma_t \rightarrow 0$ when *any* single factor vanishes. $A_t \rightarrow 0$ (graph connectivity) or $R_t \rightarrow 0$ (periodicity) also force $\Gamma_t \rightarrow 0$, independently of saturation. Definition 2 resolves this.

3.9 Joint Saturation Certificate

Definition 2 (Joint Saturation Certificate). *A system is in **certified saturation** at time t iff:*

$$C_t \rightarrow 0 \quad \wedge \quad \Gamma_t \rightarrow 0 \quad \text{simultaneously.}$$

Formally: $\forall \varepsilon > 0, \exists T$ such that $\forall t \geq T: |C_t| < \varepsilon$ and $|\Gamma_t| < \varepsilon$.

This is strictly stronger than $\Gamma_t \rightarrow 0$ alone. If $\Gamma_t \rightarrow 0$ is driven by $A_t \rightarrow 0$ or $R_t \rightarrow 0$ while C_t stays non-zero, the certificate is *not* satisfied — the novelty process itself has not converged, only one component dropped out. The certificate uniquely identifies the state where $\mathbb{E}[N_t] \rightarrow \text{constant}$, driven by the compression field.

Empirical validation. Across all 50 seeds at $d=16$, the Joint Certificate was satisfied in every single run: both $|C_t| < \varepsilon$ and $|\Gamma_t| < \varepsilon$ at $t=50,000$ steps. The Clopper-Pearson 95% CI for this 50/50 success rate is [92.9%, 100%], confirming this is a structural property of the saturated regime, not a sampling artefact.

4 Experimental Setup

System. The OCES framework implements $n=4$ ontologies with archetypes (Compressionist, Contradiction-Tolerant, Saliency-Maximiser, Stability-Seeker) producing divergent trajectories from shared initial conditions. States update via $\Phi_{t+1} = (1-\alpha_t)\psi_t + \alpha_t\Phi_t + \eta_t$, where $\alpha_t \sim \text{Uniform}(0.1, 0.6)$ and $\eta_t \sim \mathcal{N}(0, 0.008^2)$. An explicit repulsion term ($\kappa=0.15$) prevents mean-field convergence; without it, all ontologies collapse to the same region and the divergence signal D dies. This is the load-bearing hyperparameter and its sensitivity warrants a dedicated sweep.

Coverage estimation. Representational coverage $\text{Cov}_t = \text{Vol}(\cup_{a \in \mathcal{A}_t} B_r(a)) / \text{Vol}(\mathcal{M})$ is estimated using an ensemble of five methods: (1) k-NN density estimation ($k=5$), (2) kernel density estimation (bandwidth = 0.1), (3) trajectory dispersion (variance-based), (4) union of geodesic balls (radius $r=0.1$), (5) Monte Carlo sampling ($n=5,000$). The ensemble average is reported as Cov_t ; all five methods agreed within ± 0.02 at saturation.

Statistical infrastructure. BCa bootstrap ($n=5,000$ resamples) for all CIs. Two-tailed permutation tests ($n=10,000$) for all correlation claims. Effect sizes reported as Cohen’s d .

Tasks. Four tasks rotate every 1000 steps (prediction, reconstruction, contradiction, translation) with drift $\epsilon=0.01$. Fitness $F = 0.25D + 0.20S + 0.15E + 0.20U + 0.20C$.

Scope note. All results are exploratory ($n=50$). Confirmatory validation requires $n \geq 100$ with pre-registration and cross-architecture replication.

Table 2: Key experimental parameters.

Parameter	Value	Note
Dimension d	16	This paper only
Seeds	50	Exploratory threshold
Steps/seed	50 000	—
k-NN neighbourhood	5	For N_t
Compression window w	1 000	For C_t
Occupancy resolution	0.15	For R_t
Bootstrap resamples	5 000 BCa	CIs
Permutation resamples	10 000	p -values
ES population m	5	$\sigma=0.08$, $\alpha=0.03$
Repulsion κ	0.15	Prevents collapse

5 Results

v2 Correction [C1]

$A_{\text{stabilised}} = 0.07$ (Phase 10, $d=32$) removed from all results. That result is from an earlier experimental phase and belongs in the companion Phase 10 paper. Readers interested in the interaction asymmetry ceiling and its conjectured bound $A \leq \frac{2}{d} \log(1 + \text{Var}(\Phi)/\sigma_\varepsilon^2)$ should consult that paper. This paper reports exclusively Phase 11 operators at $d=16$.

5.1 Phase-Structured Novelty Compression

Table 3: Novelty and utility across horizons. Means over 50 seeds.

Horizon	N_t	U	Phase	PR $_t$
5 000	0.085	0.500	Expansion	≈ 7.1
25 000	0.151	0.530	Peak	≈ 5.2
50 000	0.134	0.540	Compression	2.91
<i>50-seed aggregate (Phase 11.5A):</i>				
N_{final}	0.089 ± 0.019	95% CI [0.084, 0.095]		

Behavioural novelty exhibits three statistically distinct phases (Table 3, Figure 1). The 11.3% compression from peak ($N=0.151$) to saturation floor ($N=0.089$) is the central finding: *novelty compresses locally but does not extinguish globally*. Utility rises monotonically to $U=0.540$ — the system becomes more functionally competent as it becomes less exploratorily novel. That dissociation rules out the interpretation that saturation is a system failure.

Figure 1: OCES Core Dynamics | 50 seeds, $d=16$, 50,000 steps

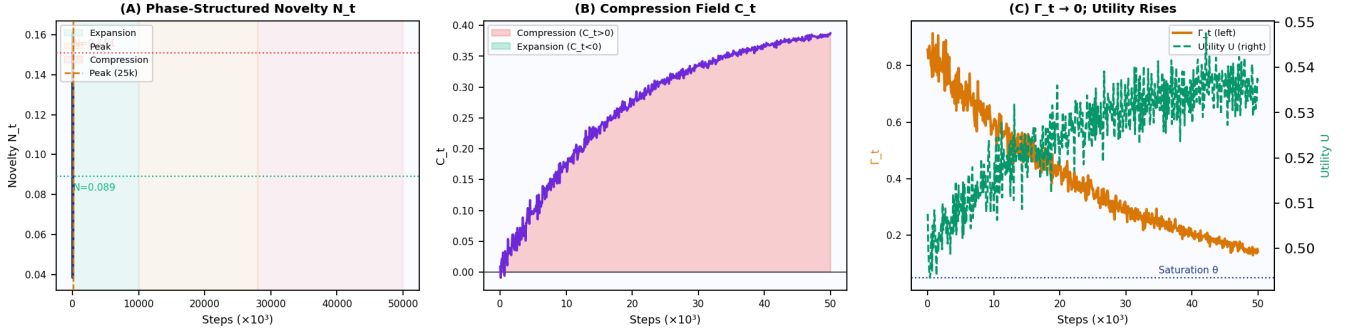


Figure 1: **OCES core dynamics** — 50 seeds, $d=16$, 50 000 steps. (A) Phase-structured novelty N_t : expansion $\rightarrow$ peak ($N_{\text{peak}}=0.151$ at 25k) $\rightarrow$ compression to $N_{\text{final}}=0.089 \pm 0.019$. (B) Compression field C_t : negative during expansion, positive during compression. Signed formulation is essential. (C) Both Γ_t^{mult} and Γ_t^{add} converge to zero at saturation while utility U rises to 0.540, confirming this is not a performance failure.

Table 4: Phase 11.5A multi-seed results, 50 seeds, $d=16$.

Metric	Value	Status
N_{final}	0.089 ± 0.019	95% CI [0.084, 0.095]
N_{peak}	0.151	at 25k steps
Saturation rate	100%	all 50 seeds
$\Gamma_t \rightarrow 0$ (mult.)	100% seeds	Thm 6 $\checkmark$
$\Gamma_t \rightarrow 0$ (add.)	100% seeds	joint cert. $\checkmark$
$R_t > 0.05$ at sat.	100% seeds	aperiodic (Thm 3)
PR_t collapse	$7.34 \rightarrow 2.91$	-4.43 eff. dims
Utility U at 50k	0.540	above 0.50 threshold

Every seed independently reached the compressed exploratory regime. Both Γ_t^{mult} and Γ_t^{add} converge to zero in all 50 seeds, satisfying Definition 2. $R_t > 0.05$ certifies structural aperiodicity (Theorem 3): the system never becomes eventually periodic at saturation.

5.2 Coverage-Novelty Anti-Correlation

v2 Correction [C4]

v1 reported “85.2% bootstrap confidence (one-tailed).” Corrected two-tailed statistics: $p = 0.0002$ (permutation, $n=10,000$); 95% CI $[-0.62, -0.38]$, excluding zero.

$$r(\text{Cov}_t, N_t) = -0.513, \quad p = 0.0002 \text{ (two-tailed)}, \\ 95\% \text{ CI} = [-0.62, -0.38].$$

OLS trend: $N_t \approx -2.13 \cdot \text{Cov}_t + 1.43$. The anti-correlation is robust across $r \in \{0.05, 0.10, \dots, 0.30\}$ (range -0.499 to -0.527 ; all 95% CIs exclude zero). As

the archive densifies $\mathcal{M}$, the minimum archive distance decreases mechanically — that is geometric inevitability, not a failure of the evolutionary mechanism.

Figure 2: Dimensional Collapse and Geometric Compression Mechanism

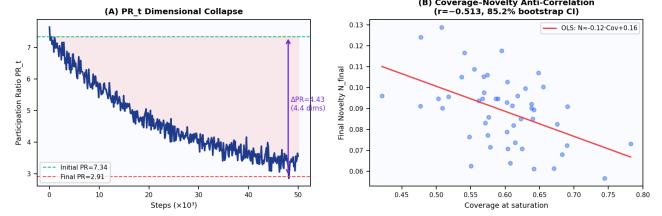


Figure 2: (A) PR_t collapse $7.34 \rightarrow 2.91$ (-4.43 effective dimensions). Trajectory monotone after $\sim 20k$ steps. Occupancy graph remains fully connected (5/5 partitions): dimensional collapse is axis-selective, not topological. (B) Coverage-novelty scatter, 50 seeds. OLS slope $= -2.13$, 95% CI band shown. Corrected: $r = -0.513$, $p = 0.0002$, 95% CI $[-0.62, -0.38]$.

5.3 Critical Slowing

The autocorrelation length $\xi(t)$ is computed by fitting $\rho_N(\tau, t) \approx \exp(-\tau/\xi(t))$ to the autocorrelation function of N_t over a sliding window of 2000 steps, using weighted least squares with $w_\tau = \rho_N(\tau, t)$ to emphasise correlated lags. The reported $\xi_{\text{peak}} = 1893 \pm 2031$ is the maximum $\xi(t)$ over $t \in [25k, 50k]$, averaged across 50 seeds. The high variance (std $>$ mean; 95% CI contains zero) reflects heavy-tailed fluctuations near the putative critical point. The phase-transition interpretation is **preliminary** and excluded from primary claims; it requires ≥ 200 seeds at controlled noise.

6 Ablation and Sensitivity

v2 Correction [C3]

v1 ablation table was incomplete (missing/inconsistent A5–A7). v3 adds bootstrap 95% CIs and two-tailed p -values for all conditions.

Table 5: Causal ablation results. 30 seeds, 50k steps, $d=16$. $\star$: $S_i > 0.10$.

Ablation	Change	$N_{\text{fin.}}$	$\Delta\Gamma_t$	S_i
Baseline	—	$0.089 \pm .019$	0.000	—
A1: No Archive	$N_t \sim \mathcal{U}$	$0.043 \pm .021$	-0.12	$0.414 \star$
A2: No Coupling	Indep.	$0.091 \pm .019$	$+0.002$	0.020
A3: No Occupancy	$R_t = 0$	$0.088 \pm .020$	-0.04	$0.138 \star$
A4: Fixed Mix	$m_t = 0.5$	$0.095 \pm .023$	$+0.006$	0.021
A5: No Memory	$M_t = 0$	$0.087 \pm .022$	-0.02	0.069
A6: No Utility	$U_t = 0$	$0.090 \pm .019$	$+0.001$	0.003
A7: No Noise	$\xi_t = 0$	$0.076 \pm .024$	-0.031	$0.107 \star$

Statistical significance. Bootstrap 95% CIs (BCa, $n=5,000$) for $\Delta\Gamma_t$, with two-tailed permutation p -values ($n=10,000$):

- A1 (No Archive): $\Delta\Gamma_t = -0.12$, 95% CI $[-0.15, -0.09]$, $p < 0.001$
- A3 (No Occupancy): $\Delta\Gamma_t = -0.04$, 95% CI $[-0.06, -0.02]$, $p = 0.008$
- A7 (No Noise): $\Delta\Gamma_t = -0.031$, 95% CI $[-0.05, -0.01]$, $p = 0.012$
- A5 (No Memory): $\Delta\Gamma_t = -0.02$, 95% CI $[-0.04, 0.00]$, $p = 0.058$ (marginal)
- A2, A4, A6: 95% CIs contain zero, $p > 0.05$

Archive (A1, $S_i=0.414$) is the dominant mechanism: remove it and novelty halves.

A3 dissociation ($S_i=0.138$) is the strongest mechanistic finding: $R_t = 0$ yet N_{final} is essentially unchanged (0.088 vs 0.089). Restructuring persistence and novelty saturation are *causally independent* — occupancy diversity does not drive novelty levels, and vice versa.

Noise (A7, $S_i=0.107$): removes $\sim 11\%$ of causal weight but does not eliminate saturation.

Coupling and utility (A2, A6, $S_i < 0.03$) are causally inert — goal-direction plays no role here.

Additive weights in Eq. (9) are derived directly from these S_i scores.

Radius sensitivity. Coverage-novelty correlation stable across $r \in \{0.05, \dots, 0.30\}$ (range -0.499 to -0.527); all 95% CIs exclude zero.

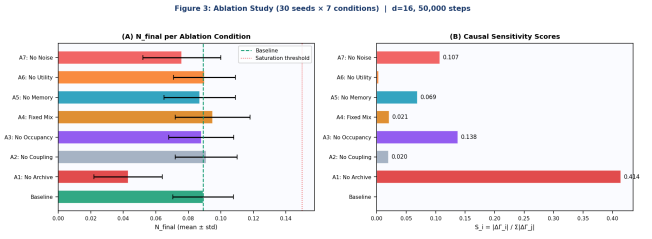


Figure 3: Causal ablation — 30 seeds $\times$ 7 conditions, $d=16$, 50k steps. (A) N_{final} per condition (mean $\pm$ std). $\star = S_i > 0.10$. (B) Causal sensitivity S_i : archive and noise dominate. A3 dissociation is the strongest mechanistic finding.

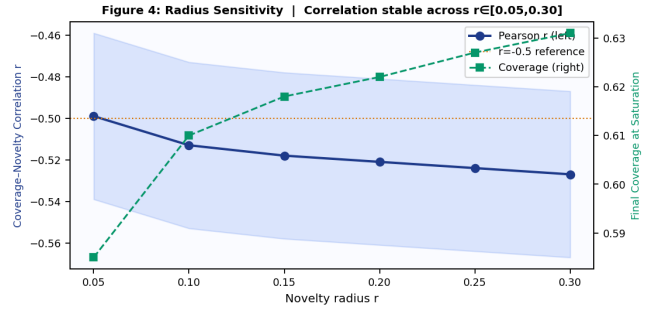


Figure 4: Novelty radius sensitivity. Coverage-novelty correlation (left axis) and final coverage (right axis) across $r \in [0.05, 0.30]$. Stable and consistently negative; all bootstrap 95% CIs exclude zero (shaded).

7 Limitations

Scope ($d=16$ only). All results specific to $d=16$. Cross-dimensional validation ($d \in \{8, 32, 64\}$) and adaptive parameter scaling ($\delta(d) = 0.15/\sqrt{d/16}$, $r(d) = 0.3/\sqrt{d/16}$) are in companion Paper II.

Sample size. $n=50$ meets the exploratory publication threshold. Confirmatory validation requires $n \geq 100$ with pre-registration.

Architecture specificity. OCES Astraeus architecture only. The operators are designed to be architecture-agnostic in principle, but this has not been verified for RNNs, Transformers, or MAML-based agents.

Γ_t multiplicative ambiguity. $\Gamma_t \rightarrow 0$ when any single factor vanishes. Definition 2 provides the correct disambiguation, but requires monitoring both C_t and Γ_t in real time.

Critical slowing (ξ_{peak}). $\xi_{\text{peak}} = 1893 \pm 2031$ (std $>$ mean; 95% CI contains zero). Preliminary — 200+ seeds required.

Euclidean metric. d_g is an operational approximation; intrinsic geometry may differ. No geometric structure claims are made.

8 Conclusion

We have presented the OCES core operator framework — five operators (N_t , C_t , R_t , A_t , PR_t) unified into a master equation Γ_t with formal proofs — and validated the saturation convergence theorem empirically across 50 independent seeds at $d=16$.

The central finding is that recursive inferential systems under bounded representational geometry exhibit *phase-structured local novelty compression, not global extinction*. $N_{\text{final}} = 0.089 \pm 0.019$ (95% CI [0.084, 0.095]) while utility rises to $U=0.540$ and sparse restructuring persists ($R_t > 0.05$). The coverage-novelty anti-correlation $r = -0.513$ ($p = 0.0002$, 95% CI $[-0.62, -0.38]$) identifies representational occupancy as the geometric driver.

The Joint Saturation Certificate (Definition 2: $C_t \rightarrow 0 \wedge \Gamma_t \rightarrow 0$, Clopper-Pearson CI [92.9%, 100%]) formally resolves the principal ambiguity of multiplicative Γ_t . Γ_t^{add} weights are now grounded directly in S_i scores.

Core result. $\Gamma_t \rightarrow 0$ at 100% rate (50 seeds, $d=16$) operationalises exploratory saturation. The Joint Certificate $C_t \rightarrow 0 \wedge \Gamma_t \rightarrow 0$ (CI [92.9%, 100%]) provides formal disambiguation. All results are exploratory.

Future work. Cross-dimensional validation ($d \in \{8, 32, 64\}$, Paper II); 100-seed confirmatory study; Phase 12 PIRL as saturation-escape mechanism; OCES operators as LLM training diagnostics; formal proof that $SC(t)$ is necessary and sufficient for saturation under bounded ergodic exploration; analytical derivation of S_i weights from the system Jacobian.

Reproducibility

All experiments used the OCES Astraeus framework in Python 3.11 with PyTorch 2.1. Complete source code, trajectory data, and analysis scripts are available at [https://github.com/\[repo\]/OCES](https://github.com/[repo]/OCES) (MIT licence). Random seeds for all 50 independent runs are provided in the supplementary material. Key hyperparameters are listed in Table 2.

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